

1. Administrative & Study Identification

Full Study Title: A Study of Siremadlin in Combination With Venetoclax Plus Azacitidine in Adult Participants With Acute Myeloid Leukemia (AML) Who Are Ineligible for Chemotherapy. **Short Title/Acronym:** Siremadlin + Venetoclax + Azacitidine in AML **Protocol Number:** CHDM201I12201 **NCT Number:** NCT05155709 **Posting ID:** 1027377931985192537 **Sponsor/Company Name:** Novartis **Investigational Product:** Siremadlin (HDM201) in combination with Venetoclax and Azacitidine (Trade names: Vidaza, Onureg; ATC Code: L01BC07) **Phase of Development:** PHASE1 (Phase Ib/II) **Medical Monitor:** Novartis Clinical Operations / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To rule out excessive toxicity and determine the Recommended Phase 2 Dose (RP2D) of siremadlin combined with venetoclax plus azacitidine (Part 1 / Safety Run-in), and to evaluate the preliminary efficacy in achieving Complete Remission (CR) in participants (Part 2 / Expansion). **Secondary Objectives:** To evaluate Overall Survival (OS), Progression-Free Survival (PFS), composite Complete Remission (CRc), and the pharmacokinetic (PK) profile of the combination therapy.

2.2 Background & Rationale Patients with AML who are ineligible for intensive chemotherapy have limited therapeutic options. While the addition of the BCL-2 inhibitor venetoclax to standard-of-care hypomethylating agents (HMAs) improves remission rates and survival, relapse rates remain high and many patients fail to achieve deep remissions (CR/CRi). Siremadlin is a selective inhibitor of the p53-MDM2 interaction. Adding siremadlin to venetoclax plus azacitidine was hypothesized to restore p53 tumor suppressor effects, overcome resistance mechanisms, and enhance the depth and durability of clinical responses in unfit AML patients.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Dose-confirmation, proof-of-concept multi-cohort trial) **Blinding:** Open-label **Randomization:** N/A (Non-randomized) **Trial Status:** TERMINATED (Study was terminated early)

3.2 Planned Enrollment & Duration Targeted Enrollment: 14 (Initially planned for ~56 participants; 9-15 per arm in Part 1;

~26 in Part 2. The study was permanently halted and terminated early by the sponsor after enrolling 14 patients to terminate the siremadlin program; this decision was not driven by safety concerns). **Number of Sites:** 27 sites open for recruitment. **Estimated Study Start:** 2022 **Estimated Primary Completion:** Study terminated prematurely.

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years at the date of signing the informed consent form (ICF). 2. Diagnosed with AML based on WHO 2016 classification and considered ineligible for standard intensive induction chemotherapy. 3. **Arm 1:** Received at least 2 cycles and not more than 4 cycles of first-line venetoclax plus azacitidine treatment and have not achieved a CR, CRi, CRh, or MLFS. **Arm 2:** Newly diagnosed AML with adverse genetic risk stratification (except TP53 mutation positive). 4. Participant must be considered ineligible for standard of care intensive induction chemotherapy defined by: ≥ 75 years of age; OR 18 to 74 years of age with at least one co-morbidity (ECOG Performance Status of 2 or 3; Cardiac history of CHF requiring treatment, LVEF $\leq 50\%$, or chronic stable angina; DLCO $\leq 65\%$ or FEV1 $\leq 65\%$). 5. ECOG Performance Status of 0-2 (if ≥ 75 years) or 0-3 (if 18-74 years). 6. WBC $< 25 \times 10^9/L$. 7. AST and ALT $\leq 3 \times ULN$. 8. Estimated Glomerular Filtration Rate (eGFR) ≥ 60 mL/min/1.73 m².

4.2 Exclusion Criteria 1. Prior exposure to MDM2-inhibitor therapy at any time. 2. Participants with TP53 mutation positive or del17p abnormalities. 3. Participants with AML-M3 / APL (Acute promyelocytic leukemia) with PML-RARA or AML secondary to Down's syndrome. 4. Participants treated with FLT3 inhibitors for AML indication. 5. Requirement for treatment with moderate or strong CYP3A4 inducers within 14 days prior to starting study treatment, or expected to receive them during the entire study. 6. Requirement for treatment with substrates of CYP3A4/5 with a narrow therapeutic index.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - **Siremadlin:** Administered orally per dose-escalation levels. - **Venetoclax:** 400 mg orally once daily following a standard 3-day ramp-up phase. - **Azacitidine (Vidaza, Onureg; ATC: L01BC07):** 75 mg/m² administered intravenously (IV) or subcutaneously (SC) on Days 1-7 or Days 1-5 and Days 8-9. **Route of Administration:** Oral (Siremadlin, Venetoclax) and IV/SC (Azacitidine). **Duration of Treatment:** Administered in 28-day cycles; treatment continues until disease progression, relapse, or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for AML cytopenias and infections. **Prohibited:** Strong/moderate CYP3A4 inducers, CYP3A4/5 substrates with narrow therapeutic indices, and concurrent FLT3 inhibitors or experimental therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Day -28 to 0	Informed Consent, Medical History, Bone Marrow Aspirate/Biopsy, Genetics (TP53 status), Labs
2	Days 1-28	Venetoclax ramp-up, Azacitidine administration, Siremadlin dosing, PK/PD sampling, Safety monitoring
3	Days 1-28	Continuous Venetoclax, scheduled Azacitidine/Siremadlin, Efficacy Assessments, AE monitoring
4	+ 30 Days	Final Safety Evaluation, Exit Interview, Survival Follow-up initiation

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- 1. Dose-Limiting Toxicities (DLTs):** Percentage of participants with DLTs as per investigator assessment reported during the first cycle (separately in Arm 1 & Arm 2) to determine the recommended dose for expansion.
- 2. Complete Remission (CR) - Arm 1:** Percentage of participants treated at the recommended dose for expansion achieving a CR as per investigator assessment. *(Note: This endpoint will not be analyzed since the recommended dose was not determined due to early termination).*

7.2 Secondary & Exploratory Endpoints

- 1. Complete Remission (CR) - Arm 2:** Percentage of participants treated at the recommended dose for expansion achieving CR.
- 2. Duration of Remission:** Time of the date of the first documented CR to the date of the first documented relapse or death due to any cause, whichever occurs first (Arm 1 and Arm 2 separately).
- 3. Composite Remission Rates:** Percentage of participants achieving CR, Complete Remission with partial hematological recovery (CRh), or Complete Remission with incomplete hematological recovery (CRi) (Arm 1 and Arm 2).
- 4. Pharmacokinetics:** PK profile of Siremadlin and Venetoclax.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection of Treatment-Emergent Adverse Events (TEAEs) tracked by CTCAE criteria (specifically monitoring for GI toxicities and myelosuppression).
Serious Adverse Events (SAEs): Reported within 24 hours of site awareness.
Data Monitoring Committee (DMC): Internal safety

review committee tasked with evaluating Part 1 dose-escalation toxicity prior to expanding to Part 2.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all enrolled participants who received at least one dose of study treatment); Intent-to-Treat (ITT) population for efficacy outcomes. **Sample Size**

Justification: A planned cohort of up to ~56 patients was statistically powered to test dose safety thresholds (mTPI/rules-based escalation) and detect early clinical efficacy signals in the expansion phase. (Actual N=14). **Handling of Missing Data:** Standard survival analysis algorithms (e.g., Kaplan-Meier method) incorporating censoring for incomplete follow-up data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Standard institutional monitoring of eCRFs and source data verification. **Audit Trail:** Validated electronic data capture with comprehensive locking and audit trail capabilities.

11. Study Identifiers & Governance

Primary ID: {{CHDM201I12201}} **Posting ID:** {{1027377931985192537}}

Registry Number: {{NCT05155709}} **Sponsor/Lead Organization:**

{{Novartis}} **Study Title:** {{A Study of Siremadlin in Combination With Venetoclax Plus Azacitidine in Adult Participants With Acute Myeloid Leukemia (AML) Who Are Ineligible for Chemotherapy.}}

Official Regulatory Title: {{A Phase Ib/II Open Label Dose Confirmation, Proof of Concept Study of Siremadlin in Combination With Venetoclax Plus Azacitidine in Unfit Adult AML Participants Who Responded Sub-optimally to First-line Venetoclax Plus Azacitidine Treatment and in Participants With Newly Diagnosed Unfit AML Presenting With High-risk Clinical Features}}

12. Clinical Framework (AML Specific)

Primary Condition / Disease Areas: {{Acute Myeloid Leukemia}}

Preferred UMLS Name: {{Acute Myeloid Leukemia}} **Diagnosis**

Threshold: {{WHO 2016 classification for AML, ineligible for intensive chemotherapy}} **Medical Methodology:** {{Targeted therapy combination (MDM2 inhibitor + BCL2 inhibitor + HMA)}}

Optimization Target: {{Complete Remission (CR) and minimal residual disease clearance}}

13. Study Procedures & Methodology

Management Pathway: {{Targeted combination pathway for relapsed/refractory or high-risk newly diagnosed AML}} **Education Protocol (HCP):** {{Safety run-in dose escalation and myelosuppression management}} **Education Protocol (Patient):** {{Informed consent for investigational triplet regimen and toxicity reporting}} **Quality Audit Mechanism:** {{Dose-escalation safety review meetings}}

14. Observation & Follow-Up Schedule

Total Duration: {{Continuous in 28-day cycles until disease progression or unacceptable toxicity}} **Onsite Visit Cadence:** {{Frequent visits during Cycle 1; Day 1 of subsequent 28-day cycles}} **Remote Monitoring Frequency:** {{As clinically indicated between treatment days}} **Checklist Review Interval:** {{Cycle-by-cycle safety checks}}

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					